



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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**Class : 11****UNIT TEST -2, AUGUST - 2024**

Time Allowed : 1.30 Hours]

CHEMISTRY

[Max. Marks : 35

PART - I

I. Choose the correct answer.

5x1=5

1. The element with positive electron gain enthalpy is

- a) Hydrogen b) Sodium c) Argon d) Fluorine

2. Effective Nuclear Charge can be calculated by

- a) $Z_{\text{eff}} = Z - S$ b) $Z_{\text{eff}} = Z + S$ c) $Z_{\text{eff}} = Z^2 - S$ d) $Z_{\text{eff}} = Z + S^2$

3. Molar heat of Vapourisation of a liquid is 4.8 KJmol^{-1} . If the entropy change is $16 \text{ Jmol}^{-1}\text{K}^{-1}$, The boiling point of the liquid is

- a) 323 K b) 27°C c) 164 K d) 0.3 K

4. The Enthalpy of formation of $\text{O}_2(\text{g})$ in the standard state is -----

- a) -273 KJmol^{-1} b) $+373 \text{ KJmol}^{-1}$ c) Zero d) None of these

5. Assertion : Tertiary Carbocations are generally formed more easily than primary Carbocations.

Reason : Hyper Conjugation as well as Inductive effect due to additional Alkyl group stabilize tertiary carbonium ions.

- a) Both Assertion and Reason are true and Reason is correct explanation
b) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
c) Assertion is True but Reason is False d) Both Assertion and Reason are False

PART - II

II. Answer any three questions. Q.No. 8 is Compulsory.

3x2=6

6. Define Modern Periodic Law.

7. Give Kelvin - Planck statement of Second Law of Thermodynamics.

8. Calculate the standard heat of Formation of Propane if its Heat of Combustion is $-2220.2 \text{ KJmol}^{-1}$.The Heat of formation of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are -393.5 and $-285.8 \text{ KJ mol}^{-1}$ respectively.



9. Explain Electrometric Effect.

10. What is meant by Hyper Conjugation?

PART - III

3x3=9

III. Answer any three questions. Q.No. 12 is compulsory.

11. Explain the Pauling method for the determination of Ionic Radius.

12. How would you explain the fact that the second ionisation potential is always higher than first ionisation potential?

13. Write down the Born - Haber Cycle for the formation of CaCl_2 .

14. What are State and Path Functions? Give two examples.

15. What are Electrophiles and Nucleophiles? Give suitable examples for each.

3x5=15

PART - IV.

IV. Answer all the questions.

16. a) i) Explain the diagonal Relationship. (3)

ii) What is Screening Effect? (2)

(OR)

b) i) What are Isoelectronic Ions? Give examples. (3)

ii) List the Characteristics of Internal Energy. (2)

17. a) List the Characteristics of Gibbs Free Energy. (5)

(OR)

b) What are Spontaneous Reactions? What are the conditions for the spontaneity of a process? (5)

18. a) Give examples for the following types of Organic Reactions. (5)

i) β -Elimination

ii) Electrophilic Substitution

(OR)

b) Derive the Relationship between ΔG and K_{eq}

